In general, Database is a collection of tables that store sets of data that can be queried for use in other applications. A database management system supports the development, administration and use of database platforms. RDBMS is a type of DBMS with a row-based table structure that connects related data elements and includes functions related to Create, Read, Update and Delete operations, collectively known as CRUD.

The data in RDBMS, is stored in database objects, called Tables. A table is a collection of related data entries and it consist of rows and columns.

A field is a column in a table that is designed to maintain specific related information about every record in the table. It is a vertical entity that contains all information associated with a specific field in a table. The fields in a student table may be of the type AdmnNo, StudName, StudAge, StudClass, Place etc.

A Record is a row, which is a collection of related fields or columns that exist in a table. A record is a horizontal entity in a table which represents the details of a particular student in a student table.

12.3 Processing Skills of SQL

The various processing skills of SQL are :

1. **Data Definition Language (DDL) :** The SQL DDL provides commands for defining relation schemas (structure), deleting relations, creating indexes and modifying relation schemas.
2. **Data Manipulation Language (DML)** : The SQL DML includes commands to insert, delete, and modify tuples in the database.
3. **Embedded Data Manipulation Language :** The embedded form of SQL is used in high level programming languages.
4. **View Definition :** The SQL also includes commands for defining views of tables.
5. **Authorization :** The SQL includes commands for access rights to relations and views of tables.
6. **Integrity :** The SQL provides forms for integrity checking using condition.
7. **Transaction control :** The SQL includes commands for file transactions and control over transaction processing.

SQL-Structured Query Language is a language used for accessing databases while **MySQL** is a database management system, like SQL Server, Oracle, Informix, Postgres, etc. **MySQL** is a **RDBMS**.

Refer installing MYSQL in Annexure -1

12th Computer Science_EM Chapter 12.indd 199 22-12-2022 11:54:46